AS · Unit 1 · WBI11 Molecules, Diet, Transport and Health

Cardiovascular health

Cardiovascular disease (CVD) covers conditions of the heart and blood vessels, most of which begin with the slow build-up of fatty plaques in the arteries. This topic explains how atherosclerosis develops, which lifestyle and genetic factors raise the risk, how obesity and cholesterol are measured, and how medicines and surgery treat the disease.

1. Cardiovascular disease and what it damages

Cardiovascular disease is any disease of the heart or blood vessels that reduces the supply of blood, and so oxygen and nutrients, to the tissues. Most CVD is caused by atherosclerosis, in which plaques build up inside the walls of arteries. The plaque narrows the lumen, which raises blood pressure and reduces flow. If the artery becomes completely blocked, the tissue it supplies loses its oxygen supply and dies.

The main forms of CVD are:

- **Coronary heart disease (CHD):** narrowing of the coronary arteries that supply the heart muscle itself.
- **Angina:** chest pain caused by a temporary shortage of oxygen to the heart muscle, usually during exertion, which eases with rest.
- **Myocardial infarction (heart attack):** a coronary artery is blocked, and the heart muscle it supplies dies from lack of oxygen.
- **Stroke:** loss of brain function caused by a blocked artery (ischaemic stroke) or a burst artery (haemorrhagic stroke) supplying the brain.
- **Aneurysm:** a weakened, bulging section of an artery wall, which may rupture and cause severe internal bleeding.

2. How atherosclerosis develops

Atherosclerosis begins with damage to the endothelium, the single layer of cells lining the artery. The endothelium is damaged by high blood pressure, by chemicals in cigarette smoke, by high concentrations of LDL cholesterol in the blood, and by oxidative damage from free radicals. Damage triggers an inflammatory response. White blood cells, particularly macrophages, move into the wall of the artery and take up fats, including cholesterol from LDL. These cells become foam cells, which accumulate to form a fatty streak and then a plaque.

Over time the plaque enlarges, its cap can crack, and the artery wall loses its elasticity. The lumen narrows and blood pressure rises. A damaged plaque surface also activates platelets, which can form a clot (thrombus). A clot that grows large enough can block the artery completely.

From endothelial damage to a blocked artery

1. The endothelium is damaged, for example by high blood pressure or by smoke chemicals.
2. The damaged lining lets LDL cholesterol enter the artery wall.

   Damaged endothelium is more permeable, so LDL particles pass into the wall more easily.
3. Inflammation attracts white blood cells, which take up the cholesterol and become foam cells.
4. Foam cells and fibrous material build a plaque, which narrows the lumen.

   Narrowing raises the resistance to blood flow, so blood pressure rises further.
5. The plaque may rupture and trigger a clot, which can completely block the artery.

   Blood flow past the blockage stops, so the tissue beyond it is deprived of oxygen.

KEY IDEA

Atherosclerosis is a process that takes years. Damage to the endothelium is the starting point, and cholesterol deposition, inflammation and clot formation follow from it.

3. Risk factors

A risk factor is something that makes CVD more likely. Some risk factors cannot be changed, while others can be modified through lifestyle and treatment. Risk factors are usually linked: obesity, inactivity and high-fat diets often occur together and push blood pressure and cholesterol in the same direction.

Non-modifiable	Modifiable
Age	Diet
Gender	Smoking
Family history and genes	High blood pressure
	High blood cholesterol (especially LDL)
	Obesity and high body mass index
	Physical inactivity
	Stress
	Excess alcohol

Age and gender

The risk of CVD increases with age in both men and women. With age, arteries lose elasticity and become stiffer, so the pressure needed to push blood through them rises. Men have a higher risk than women at younger ages.

Oestrogen, present in women before the menopause, helps keep LDL low and HDL high, and it reduces plaque build-up. After the menopause, the protective effect falls and the difference between the sexes narrows.

Diet and cholesterol

Dietary fat affects blood cholesterol. Saturated fats, found in fatty meat and full-fat dairy, raise LDL. Unsaturated fats, such as vegetable oil, tend to lower LDL and contribute to HDL. Fish oil contains omega-3 fatty acids, which are associated with a lower risk of heart disease. Salt raises blood pressure because sodium retains water in the blood, increasing blood volume and the pressure it exerts on the artery walls. Fruit and vegetables contain antioxidants, such as vitamin C and polyphenols, which neutralise free radicals and protect the endothelium from oxidative damage.

EXAM TIP

When you discuss diet, say which type of fat, which lipoprotein it affects and how that affects the artery. A general statement that "fat is bad" will not earn full marks.

LDL and HDL

Cholesterol is carried in the blood by lipoproteins, which are particles of protein and lipid. Low-density lipoprotein (LDL) carries cholesterol from the liver to the tissues, and excess LDL deposits cholesterol in artery walls. High-density lipoprotein (HDL) carries cholesterol from the tissues back to the liver for breakdown. High LDL and low HDL together increase the risk of atherosclerosis.

Feature	LDL	HDL
Main role	Delivers cholesterol to tissues	Removes cholesterol from tissues to the liver
Effect on arteries	Promotes plaque formation	Helps reduce plaque
Linked fat type	Saturated fats	Unsaturated fats
Influenced by	Saturated fat intake, high blood cholesterol	Exercise and moderate alcohol raise it; smoking lowers it

Smoking

Cigarette smoke affects the cardiovascular system in several ways. Carbon monoxide binds to haemoglobin to form carboxyhaemoglobin, which is more stable than oxyhaemoglobin. This reduces the oxygen carried by the blood, so the heart and brain receive less oxygen and must increase heart rate to compensate. Nicotine stimulates the release of adrenaline, which raises heart rate and constricts arteries, raising blood pressure. Smoking also lowers HDL, promotes platelet activation and clot formation, and increases free radicals that damage the endothelium. Together these effects make atherosclerosis and thrombosis more likely.

How smoking raises the risk of CVD

1. Carbon monoxide forms carboxyhaemoglobin, which carries less oxygen.

   Less oxygen reaches the heart and brain, so they work harder to compensate.

2. Nicotine releases adrenaline, which raises heart rate and constricts arteries.

   Constricted arteries have higher resistance, so blood pressure rises.

3. Smoking lowers HDL and increases free radicals.

   Less HDL means less cholesterol is removed from the artery walls, and free radicals damage the endothelium.

4. Platelets become more active, so clots form more readily.

   A clot on a damaged plaque can block the artery and cause a heart attack or stroke.

Exercise, stress and alcohol

Physical activity lowers CVD risk. Exercise strengthens the heart muscle so that it contracts more forcefully and pumps more blood per beat, which lowers resting heart rate. It also dilates blood vessels, which lowers blood pressure, raises HDL and helps control weight. Exercise uses energy from fat as well as carbohydrate, so it helps balance energy intake and output.

Stress raises blood pressure and heart rate in the short term, and in some people it leads to overeating and obesity, which adds to the risk. Moderate alcohol intake is linked to higher HDL, and red wine contains antioxidants. However, heavy drinking raises blood pressure, contributes to obesity and irregular heartbeats, and damages the liver. Safe limits are around 3 to 4 units a day for men and 2 to 3 units for women.

4. Measuring obesity

Obesity is excess body fat, and it is a major risk factor for CVD because excess fat raises blood pressure and cholesterol. Two simple measurements are used to assess it.

Body mass index (BMI)

BMI is a quick way to check whether a person's weight is appropriate for their height. It is calculated by dividing body mass in kilograms by height in metres squared. Obesity-related health problems rise when BMI exceeds about 25, and a BMI of 30 or more is classed as obese.

Worked example: BMI for 95 kg and 1.70 m

1. Write the formula: BMI = mass ÷ height², with mass in kg and height in m.

2. Square the height: 1.70 × 1.70 = 2.89 m².

   Height is squared because body volume scales with the square of linear dimensions, so this corrects for taller people being heavier.

3. Divide the mass by the squared height: 95 ÷ 2.89 = 32.87.

4. Compare with the table: 32.87 lies between 30 and 40, so the person is classed as obese.

BMI (kg m^{-2})	Classification
Below 20	Underweight
20 to 24.9	Normal
25 to 29.9	Overweight
30 to 40	Obese
Above 40	Severely obese

COMMON MISTAKE

BMI does not distinguish fat from muscle. Athletes can have a high BMI with little body fat, and older people can have a normal BMI while losing muscle and gaining fat. It is also not reliable for children and pregnant women.

Body mass changes when the energy taken in as food and the energy used by the body are out of balance.

Waist-to-hip ratio

Fat stored around the abdomen, sometimes called central or android fat, is more closely linked to heart disease than fat on the hips and thighs. The waist-to-hip ratio measures this distribution. It is calculated by dividing the waist measurement by the hip measurement. A ratio above 0.80 for women or 0.95 for men indicates a raised risk of several diseases.

5. Treatment of cardiovascular disease

Treatment is chosen according to the severity of the disease and the risk factors present. It usually begins with lifestyle changes, and medicines or procedures are added when these are not enough. The aims are to lower blood pressure, reduce cholesterol, prevent clots and restore blood flow.

Lifestyle changes

Giving up smoking, eating less saturated fat and salt, eating more fruit and vegetables, taking regular exercise, keeping body weight in a healthy range and drinking alcohol within safe limits all reduce risk. These changes are the foundation of treatment because they target the causes rather than the symptoms.

Medicines that lower blood pressure

- **ACE inhibitors:** block angiotensin-converting enzyme, so less angiotensin II is made. Less angiotensin II means less vasoconstriction, so blood pressure falls. Side effects include a dry cough and dizziness.
- **Angiotensin receptor blockers:** block the receptors that angiotensin II acts on, so the arteries do not constrict in response. Side effects include dizziness and rash.
- **Beta blockers:** block adrenaline from binding to beta receptors in the heart. This slows the heart rate and reduces the force of contraction, so the output per minute falls and blood pressure drops.
- **Diuretics:** act on the kidneys to increase the volume of urine, removing excess salt and water. Blood volume falls, so less blood is pumped and pressure drops.
- **Calcium channel blockers:** stop calcium entering the cells of the heart and the walls of arteries. The arteries relax, and the heart works less hard.

Medicines that lower cholesterol or prevent clots

Statins are drugs that inhibit HMG-CoA reductase, the enzyme in the liver that makes cholesterol. Less cholesterol is made, so the liver takes more LDL from the blood, and LDL levels fall. Anticoagulants such as warfarin and heparin interfere with the clotting cascade, reducing the production of clotting factors. Antiplatelet drugs such as aspirin make platelets less sticky, so they are less likely to aggregate and form clots. These drugs reduce the risk of a clot forming on a plaque, but they increase the risk of bleeding, so the balance of benefit and risk must be assessed for each patient.

Surgery and procedures

When an artery is severely narrowed, surgery can restore blood flow. In balloon angioplasty, a catheter with a deflated balloon is threaded into the narrowed coronary artery. The balloon is inflated to push the plaque against the wall, and a stent, a small mesh tube, is often left in place to keep the artery open. In coronary artery bypass grafting, a healthy vessel taken from another part of the body is joined above and below the blockage, so blood flows around it.

KEY IDEA

Drugs that lower blood pressure reduce the load on the heart and the risk of artery damage. Drugs that lower cholesterol or prevent clots reduce the risk of blockage. Surgery restores flow once the blockage already exists.

Quick checks

QUICK CHECK

Which step is the earliest in the development of atherosclerosis?

- Formation of a blood clot on the plaque
- Narrowing of the lumen by fatty plaque
- Damage to the endothelium
- Rupture of an aneurysm

Atherosclerosis begins when the endothelial lining is damaged, which allows cholesterol and white blood cells into the wall. Clots and narrowing come later. Students often pick narrowing, because it is the most visible result, but it is not the first step.

QUICK CHECK

A person has a mass of 95 kg and a height of 1.70 m. What is their BMI?

- 55.8
- 25.9
- 32.9
- 0.56

BMI = mass ÷ height², so 95 ÷ (1.70 × 1.70) = 95 ÷ 2.89 = 32.87, which rounds to 32.9 and falls in the obese range. The wrong answers come from dividing by height rather than height squared, or by dividing in the wrong order.

QUICK CHECK

Why does smoking raise blood pressure?

- Carbon monoxide makes the blood thicker
- Nicotine makes the heart beat more slowly
- Nicotine releases adrenaline, which raises heart rate and constricts arteries
- Smoke causes the kidneys to hold water

Nicotine stimulates adrenaline secretion, increasing heart rate and narrowing arteries. Both raise blood pressure. Carbon monoxide reduces the oxygen the blood carries rather than making it thicker, which is a common confusion.

QUICK CHECK

Which drug type works by stopping the liver from making cholesterol?

- Beta blockers
- Diuretics
- Statins
- Aspirin

Statins inhibit HMG-CoA reductase, the enzyme that makes cholesterol in the liver. Beta blockers lower heart rate, diuretics remove water and salt, and aspirin reduces platelet stickiness. Those three work on different targets, which is why they are often confused.